

Note on Bandit Algorithms

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1 Bandits, Probability and Concentration

1.1 Introduction

Definition.

- Actions : $A_1, A_2, \dots \in \mathcal{A}$ and we call $\mathcal{A}$ the action set.
- Rewards : $X_1, X_2, \dots \in \mathcal{R}$.
- History : $H_{t-1} = (A_1, X_1, \dots, A_{t-1}, X_{t-1})$.
- Policy : $\pi \in \Pi : H_{t-1} \mapsto A_t$ and we call Π the competitor class.
- Environment: $\mathcal{E} : A_t \mapsto X_t$.
- Regret of a learner relative to a policy π until $n \in \mathbb{N}$ is

$$R_n = \mathbb{E} \left[\sum_{t=1}^n \mathcal{E}(\pi(H_{t-1})) - \sum_{t=1}^n X_t \right].$$

where H_0 represents the initial condition.

Example A. n environment is called a **stochastic Bernoulli bandit** if

- $\mathcal{A} = \{1, 2, \dots, k\}$,
- $X_t \in \{0, 1\}$,
- $\forall a \in \mathcal{A}, \exists \mu_a \in [0, 1]^k$ such that $\mathbb{P}[X_t = 1 | A_t = a] = \mu_a$.

Here the optimal policy is to play fixed action $a^* = \operatorname{argmax}_{a \in \mathcal{A}} \mu_a$. That is, the competitor class Π is the set of all constant policies $\Pi = \{\pi_1, \dots, \pi_k\}$ where π_i chooses action i in every round. The regret relative to the optimal policy π^* until n rounds is

$$\begin{aligned} R_n &= \mathbb{E} \left[\sum_{t=1}^n \mathcal{E}(\pi^*(H_{t-1})) - \sum_{t=1}^n X_t \right] \\ &= n \max_{a \in \mathcal{A}} \mu_a - \mathbb{E} \left[\sum_{t=1}^n X_t \right]. \end{aligned}$$

Definition. Stochastic stationary bandits are the environments restricted to generate the rewards in response to each action from a distribution independent of the previous history.

Example Linear Stochastic Bandits.

- $\mathcal{A} \subset \mathbb{R}^d$,
- $X_t = \langle a, \theta \rangle + \eta_t$ for $\theta \in \mathbb{R}^d, a \in \mathcal{A}$ and η_t is a standard Gaussian.

1.2 Foundations of Probability

Definition. Let an event set Ω be given.

- A set $\mathcal{F} \subset 2^\Omega$ is called a **σ -algebra** on Ω if
 - (1) $\Omega \in \mathcal{F}$,
 - (2) $A \in \mathcal{F} \implies A^c \in \mathcal{F}$,
 - (3) $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^\infty A_i \in \mathcal{F}$.
- $\mathcal{G} \subseteq \mathcal{F}$ is a **sub- σ -algebra** if $\mathcal{G}$ is a σ -algebra on Ω .
- $(\Omega, \mathcal{F})$ is called a **measurable space** if $\mathcal{F}$ is a σ -algebra on Ω .
- $\mathcal{B}(\mathbb{R})$ is the **Borel σ -algebra** on $\mathbb{R}$, which is the smallest σ -algebra containing all open intervals (a, b) for $a, b \in \mathbb{R}$.

Definition. Let a measurable space $(\Omega, \mathcal{F})$ be given.

- A function $\mu : \mathcal{F} \mapsto [0, \infty]$ is **σ -additive** if
 - (1) $\mu(\emptyset) = 0$,
 - (2) $A_1, A_2, \dots \in \mathcal{F}$ are disjoint $\implies \mu(\bigcup_{i=1}^\infty A_i) = \sum_{i=1}^\infty \mu(A_i)$.
- A function $\mu : \mathcal{F} \mapsto [0, \infty]$ is a **measure** if μ is σ -additive.
- $(\Omega, \mathcal{F}, \mu)$ is called a **measure space** if μ is a measure on $(\Omega, \mathcal{F})$.
- A measure μ is a **probability measure** if $\mu(\Omega) = 1$.
- $(\Omega, \mathcal{F}, \mu)$ is called a **probability space** if μ is a probability measure on $(\Omega, \mathcal{F})$.
- A function $X : \Omega \mapsto \mathbb{R}$ is called a **measurable function** if $\forall B \in \mathcal{B}(\mathbb{R}), X^{-1}(B) \in \mathcal{F}$.
- A **filtration** is a sequence $(\mathcal{F}_t)_{t=0}^n$ of sub- σ -algebras of $\mathcal{F}$ where $\mathcal{F}_t \subseteq \mathcal{F}_{t+1}$ for all $t < n$.

Definition. Let a measure space $(\Omega, \mathcal{F}, \mu)$ be given.

- A function $f : \Omega \mapsto [0, \infty]$ is called a **simple function** if $f = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ for some $a_1, a_2, \dots, a_n \in [0, \infty]$ and $A_1, A_2, \dots, A_n \in \mathcal{F}$.
- The **integral of a simple function** $f = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ is defined as $\int_\Omega f d\mu := \sum_{i=1}^n a_i \mu(A_i)$.
- The **integral of a non-negative measurable function** $f : \Omega \mapsto [0, \infty]$ is defined as $\int_\Omega f d\mu := \sup_g \int_\Omega g d\mu$ where the supremum is taken over all simple functions g such that $0 \leq g(\omega) \leq f(\omega)$ for all $\omega \in \Omega$.
- The **integral of a measurable function** $f : \Omega \mapsto \mathbb{R}$ is defined as $\int_\Omega f d\mu := \int_\Omega f^+ d\mu - \int_\Omega f^- d\mu$ where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$.
- A function $f : \Omega \mapsto \mathbb{R}$ is called **integrable** if $\int_\Omega |f| d\mu < \infty$.

Definition. Let a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ be given.

- A measurable function $X : \Omega \mapsto \mathbb{R}$ is called a **random variable**.
- The **conditional probability** $\mathbb{P}[A|B] := \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]}$ for $A, B \in \mathcal{F}$ and $\mathbb{P}[B] > 0$.
- Two events $A, B \in \mathcal{F}$ are called **independent** if $\mathbb{P}[A \cap B] = \mathbb{P}[A] \cdot \mathbb{P}[B]$.
- Two random variables X, Y are called **independent** if $\forall A, B \in \mathcal{B}(\mathbb{R}), \mathbb{P}[X \in A, Y \in B] = \mathbb{P}[X \in A] \cdot \mathbb{P}[Y \in B]$.

Definition. Let a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable X be given.

- The **expectation** of a random variable X is defined as $\mathbb{E}[X] := \int_{\Omega} X d\mathbb{P}$.
- For a sub- σ -algebra $\mathcal{H} \subseteq \mathcal{F}$, the **conditional expectation** of X given $\mathcal{H}$ is denoted by $\mathbb{E}[X|\mathcal{H}]$ and is defined as a random variable g on Ω such that for all $H \in \mathcal{H}$,

$$\int_H g d\mathbb{P} = \int_H X d\mathbb{P}.$$

Theorem 2.11. If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and X is an integrable random variable and $\mathcal{H} \subseteq \mathcal{F}$ is a sub- σ -algebra, there exists a unique (almost surely) random variable $\mathbb{E}[X|\mathcal{H}]$ such that for all $H \in \mathcal{H}$,

$$\int_H \mathbb{E}[X|\mathcal{H}] d\mathbb{P} = \int_H X d\mathbb{P}.$$

Theorem 2.12. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\mathcal{G}, \mathcal{G}_1, \mathcal{G}_2 \subseteq \mathcal{F}$ be sub- σ -algebras of $\mathcal{F}$, X, Y be integrable random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Then the following properties hold:

- (1) If $X \geq 0$ almost surely, then $\mathbb{E}[X|\mathcal{G}] \geq 0$ almost surely.
- (2) $\mathbb{E}[1|\mathcal{G}] = 1$ almost surely.
- (3) $\mathbb{E}[X + Y|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}] + \mathbb{E}[Y|\mathcal{G}]$ almost surely.
- (4) If Y is $\mathcal{G}$ -measurable and $\mathbb{E}[XY] < \infty$, then $\mathbb{E}[XY|\mathcal{G}] = Y\mathbb{E}[X|\mathcal{G}]$ almost surely.
- (5) If $\mathcal{G}_1 \subseteq \mathcal{G}_2$, then $\mathbb{E}[X|\mathcal{G}_1] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}_2]|\mathcal{G}_1]$ almost surely.

1.3 Stochastic Processes and Markov Chains

Definition. Let two measurable spaces $(\mathcal{X}, \mathcal{F})$ and $(\mathcal{Y}, \mathcal{G})$ be given.

- $(\mathcal{X}, \mathcal{F})$ and $(\mathcal{Y}, \mathcal{G})$ are called **isomorphic** if there exists a bijective function $f : \mathcal{X} \mapsto \mathcal{Y}$ such that $A \in \mathcal{F} \iff f(A) \in \mathcal{G}$.
- $(\mathcal{X}, \mathcal{F})$ is a **Borel space** if it is isomorphic to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ for some set $A \in \mathcal{B}(\mathbb{R})$.

Theorem 3.1. Let μ be a probability measure on a Borel space S and λ be the Lebesgue measure on $([0, 1], \mathcal{B}([0, 1]))$. Then there exists a sequence of independent random element $X_1, X_2, \dots$ on $([0, 1], \mathcal{B}([0, 1]), \lambda)$ such that $\lambda_{X_t}(\cdot) := \lambda(X_t^{-1}(\cdot)) = \mu(\cdot)$ for all t .

Definition. For "countable" set $\mathcal{T}$, A stochastic process is a collection of random variables $X_t : t \in \mathcal{T}$.

Theorem 3.2. Let Borel spaces $(\Omega_n, \mathcal{F}_n)$ and measures μ_n on $(\Omega_1 \times \dots \times \Omega_n, \mathcal{F}_1 \otimes \dots \otimes \mathcal{F}_n)$ are given for each $n \in \mathbb{N}$.

$$\begin{aligned} &\forall n \in \mathbb{N} \text{ and } A \in \mathcal{F}_1 \otimes \dots \otimes \mathcal{F}_n, \quad \mu_{n+1}(A \times \Omega_{n+1}) = \mu_n(A) \\ &\implies \exists \text{ a probability space } (\Omega, \mathcal{F}, \mathbb{P}) \\ &\text{and random elements } X_1, X_2, \dots \text{ with } X_t : \Omega \rightarrow \Omega_t \text{ such that } \mathbb{P}_{X_1, \dots, X_n} = \mu_n \text{ for all } n. \end{aligned}$$

Definition. Let $(\mathcal{X}, \mathcal{F})$ and $(\mathcal{Y}, \mathcal{G})$ be measurable spaces. A **probability kernel** or **Markov kernel** from $(\mathcal{X}, \mathcal{F})$ to $(\mathcal{Y}, \mathcal{G})$ is a function $K : \mathcal{X} \times \mathcal{G} \rightarrow [0, 1]$ such that

- (a) $K(x, \cdot)$ is a measure for all $x \in \mathcal{X}$ and
- (b) $K(\cdot, A)$ is $\mathcal{F}$ -measurable for all $A \in \mathcal{G}$.

Notation: $K_x(A)$ or $K(A|x)$

Definition. Let K_1 be a probability kernel from $(\mathcal{X}, \mathcal{F}) \rightarrow (\mathcal{Y}, \mathcal{G})$ and K_2 be a probability kernel from $(\mathcal{Y}, \mathcal{G}) \rightarrow (\mathcal{Z}, \mathcal{H})$. Then the product kernel $K_1 \otimes K_2$ is the probability kernel from $(\mathcal{X}, \mathcal{F})$ to $(\mathcal{Y} \times \mathcal{Z}, \mathcal{G} \otimes \mathcal{H})$ defined by

$$(K_1 \otimes K_2)(x, A) = \int_{\mathcal{Y}} \int_{\mathcal{Z}} \mathbb{1}_A((y, z)) K_2(y, dz) K_1(x, dy).$$

Theorem 3.3. Let $(\Omega_n, \mathcal{F}_n)_{n=1}^{\infty}$ be a sequence of measurable spaces.

K_1 : a probability measure on $(\Omega_1, \mathcal{F}_1)$,

K_n : a probability kernel from $\Pi_{t=1}^{n-1} \Omega_t$ to Ω_n for $n \geq 2$.

$\implies \exists$ a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

and random elements $(X_t)_{t=1}^{\infty}$ with $X_t : \Omega \rightarrow \Omega_t$ such that $\mathbb{P}_{X_1, \dots, X_n} = \bigotimes_{t=1}^n K_t$ for all n .

Definition. A **homogeneous Markov chain** is a sequence of random elements $(X_t)_{t=1}^{\infty}$ taking values in state space $\mathcal{S} = (\mathcal{X}, \mathcal{F})$ and with

$$\mathbb{P}(X_{t+1} \in \cdot | X_1, \dots, X_t) = \mathbb{P}(X_{t+1} \in \cdot | X_t) = \mu(X_t, \cdot), \quad \text{a.s.}$$

Here μ is a probability kernel from $(\mathcal{X}, \mathcal{F})$ to $(\mathcal{X}, \mathcal{F})$.

Definition 3.4. A $\mathbb{F}$ -adapted sequence of random variables $(X_t)_{t \in \mathbb{N}_+}$ (i.e. X_t is $\mathcal{F}_t$ -measurable for all $t \in \mathbb{N}$) is a $\mathbb{F}$ -adapted **martingale** if

(a) $\mathbb{E}[|X_t|] < \infty$ for all t .

(b) $\mathbb{E}[X_{t+1} | \mathcal{F}_t] = X_t$ for all t .

Supermartingale if less-than, submartingale if greater-than.

Example 3.5. $(Y_t)_{t \in \mathbb{N}}$ be a sequence of independent random variables with $\mathbb{P}(Y_t = 1) = \mathbb{P}(Y_t = -1) = \frac{1}{2}$. Then the winnings after t rounds is $S_t = \sum_{i=1}^t Y_i$, a martingale adapted to the filtration $(\mathcal{F}_t)_{t \in \mathbb{N}_+}$, where $\mathcal{F}_t = \sigma(Y_1, \dots, Y_t)$.

Definition 3.6. Let $\mathbb{F} = (\mathcal{F}_t)_{t \in \mathbb{N}}$ be a filtration. A random variable $\tau \in \mathbb{N} \cup \{\infty\}$ is a **stopping time** with respect to the filtration $\mathbb{F}$ if

$$\{\tau \leq t\} \in \mathcal{F}_t \text{ for all } t \in \mathbb{N}.$$

The σ -algebra generated by the stopping time τ is

$$\mathcal{F}_\tau = \{A \in \mathcal{F}_\infty : A \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for all } t \in \mathbb{N}\}.$$

Theorem 3.8 (Doob's Optional Stopping Theorem). Let $\mathbb{F} = (\mathcal{F}_t)_{t \in \mathbb{N}}$ be a filtration and $(X_t)_{t \in \mathbb{N}}$ be an $\mathbb{F}$ -adapted martingale. Let τ be a stopping time with respect to $\mathbb{F}$, and one of the following conditions is satisfied:

(a) There exists an $n \in \mathbb{N}$ such that $\mathbb{P}(\tau > n) = 0$.

(b) $\mathbb{E}[\tau] < \infty$ and there exists a constant $c \in \mathbb{R}$ such that $|X_{t+1} - X_t| \leq c$ for all t , $\mathbb{E}[|X_{t+1} - X_t| | \mathcal{F}_t] \leq c$ a.s. for $\tau > t$.

(c) There exists a constant c such that $|X_{t \wedge \tau}| \leq c$ a.s. for all $t \in \mathbb{N}$.

Then X_τ is almost surely well-defined and

$$\mathbb{E}[X_\tau] = \mathbb{E}[X_0].$$

Theorem 3.9 (Maximal inequality). Let $(X_t)_{t=0}^{\infty}$ be a supermartingale with $X_t \geq 0$ a.s. for all $t \in \mathbb{N}$. Then for any $\epsilon > 0$,

$$\mathbb{P}\left(\sup_{t \in \mathbb{N}} X_t \geq \epsilon\right) \leq \frac{\mathbb{E}[X_0]}{\epsilon}.$$

Theorem 3.10. Let $(X_t)_{t=0}^{\infty}$ be a submartingale with $X_t \geq 0$ a.s. for all $t \in \mathbb{N}$. Then for any $\epsilon > 0$,

$$\mathbb{P}\left(\max_{t \in \{0, 1, \dots, n\}} X_t \geq \epsilon\right) \leq \frac{\mathbb{E}[X_n]}{\epsilon}.$$

1.4 Stochastic Bandits

Definition. A **stochastic bandit** is a collection of distributions $\nu = \{P_a : a \in \mathcal{A}\}$.

Here we are challenging to maximize the total reward $S_n = \sum_{t=1}^n X_t$ over n rounds.

Definition. A set of bandits $\mathcal{E}$ for which $\nu \in \mathcal{E}$ is called the environment class.

- An environment class $\mathcal{E}$ is **unstructured** if $\mathcal{A}$ is finite and

$$\forall a \in \mathcal{A}, \exists \text{ sets of distributions } \mathcal{M}_a \text{ such that } \mathcal{E} = \{\nu = (P_a : a \in \mathcal{A}) : P_a \in \mathcal{M}_a, \forall a \in \mathcal{A}\}.$$

or, in short, $\mathcal{E} = \times_{a \in \mathcal{A}} \mathcal{M}_a$.

- Environment classes with finite number of degrees of freedom are called **parametric** (Bernoulli).
- Environment classes that are not parametric are called **nonparametric** (subgaussian).
- An environment class is called **structured** if it is not unstructured.

Example 4.1. Let $\mathcal{A} = \{1, 2\}$ and $\mathcal{E} = \{(\mathcal{B}(\theta), \mathcal{B}(1 - \theta)) : \theta \in [0, 1]\}$ where $\mathcal{B}(\theta)$ denotes the Bernoulli distribution with mean θ . Then $\mathcal{E}$ is a structured parametric environment class.

Example 4.2 (Stochastic linear bandit). Let $\mathcal{A} \subseteq \mathbb{R}^d$ and $\theta \in \mathbb{R}^d$ and

$$\nu_\theta = (\mathcal{N}(\langle a, \theta \rangle, 1) : a \in \mathcal{A}) \quad \text{and} \quad \mathcal{E} = \{\nu_\theta : \theta \in \mathbb{R}^d\}.$$

Example 4.3. Consider an undirected graph G with vertices $V = 1, \dots, |V|$ and edges $E = 1, \dots, |E|$. In each round the learner chooses a path from vertex 1 to vertex $|V|$. Then each edge $e \in [E]$ is removed from the graph with probability $1 - \theta_e$ for unknown $\theta \in [0, 1]^{|E|}$. In this problem, $\mathcal{A}$ is the set of paths and

$$\nu_\theta = \left(\mathcal{B} \left(\prod_{e \in a} \theta_e \right) : a \in \mathcal{A} \right) \quad \text{and} \quad \mathcal{E} = \{\nu_\theta : \theta \in [0, 1]^{|E|}\}.$$

Definition. Let $\nu = (P_a : a \in \mathcal{A})$ be a stochastic bandit.

$$\mu_a(\nu) := \int_{-\infty}^{\infty} x dP_a(x)$$

Also let $\mu^*(\nu) := \max_{a \in \mathcal{A}} \mu_a(\nu)$, the largest mean of all the arms.

Definition. The **regret** of policy π on bandit instance ν is

$$R_n(\pi, \nu) := n\mu^*(\nu) - \mathbb{E} \left[\sum_{t=1}^n X_t \right]$$

where the expectation is taken with respect to the probability measure on outcomes induced by the interaction of π and ν .

Lemma 4.4. Let ν be a stochastic bandit environment. Then

- $R_n(\pi, \nu) \geq 0$ for all policies π ,
- the policy π choosing $A_t \in \arg \max_a \mu_a(\nu)$ for all t satisfies $R_n(\pi, \nu) = 0$, and
- if $R_n(\pi, \nu) = 0$ for some policy π , then $\mathbb{P}[\mu_{A_t}(\nu) = \mu^*(\nu)] = 1$ for all $t \in [n]$.

Proof. (a) Since $\mu^*(\nu) = \max_{a \in \mathcal{A}} \mu_a(\nu)$,

$$n\mu^*(\nu) \geq \mathbb{E} \left[\sum_{t=1}^n \mu_{A_t}(\nu) \right].$$

By the linearity of expectation,

$$\mathbb{E} \left[\sum_{t=1}^n \mu_{A_t}(\nu) \right] = \sum_{t=1}^n \mathbb{E} [\mu_{A_t}(\nu)].$$

By the stochastic bandit assumption and the tower property,

$$\begin{aligned} \sum_{t=1}^n \mathbb{E} [\mu_{A_t}(\nu)] &= \sum_{t=1}^n \mathbb{E} [\mathbb{E} [X_t \mid H_{t-1}, A_t]] \\ &= \sum_{t=1}^n \mathbb{E} [X_t] \\ &= \mathbb{E} \left[\sum_{t=1}^n X_t \right], \end{aligned}$$

using linearity of expectation again. Therefore,

$$R_n(\pi, \nu) = n\mu^*(\nu) - \mathbb{E} \left[\sum_{t=1}^n X_t \right] \geq 0.$$

(b) Let π be the policy choosing $A_t \in \arg \max_a \mu_a(\nu)$ for all t . Then $\mu^*(\nu) = \mu_{A_t}(\nu)$ for all t , which implies

$$\begin{aligned} R_n(\pi, \nu) &= n\mu^*(\nu) - \mathbb{E} \left[\sum_{t=1}^n X_t \right] \\ &= \sum_{t=1}^n (\mu^*(\nu) - \mathbb{E} [\mu_{A_t}(\nu)]) \\ &= \mathbb{E} \left[\sum_{t=1}^n (\mu^*(\nu) - \mu_{A_t}(\nu)) \right] \\ &= 0 \end{aligned}$$

and therefore $R_n(\pi, \nu) = 0$.

(c) Suppose $R_n(\pi, \nu) = 0$ for some policy π . Then

$$\sum_{t=1}^n \mathbb{E} [\mu^*(\nu) - \mu_{A_t}(\nu)] = 0.$$

Since $\mu^*(\nu) - \mu_{A_t}(\nu) \geq 0$ for all t , $\mu^*(\nu) = \mu_{A_t}(\nu)$ almost surely for all $t \in \{1, \dots, n\}$. □

Remark (Factorisation of the regret). The learner cannot always achieve zero regret, so the relatively weak objective is to find a policy π with

$$\forall \nu \in \mathcal{E}, \quad \lim_{n \rightarrow \infty} \frac{R_n(\pi, \nu)}{n} = 0.$$

Or further objective, finding some constants $C > 0$ and $p < 1$ such that

$$\forall \nu \in \mathcal{E}, \quad R_n(\pi, \nu) \leq Cn^p.$$

Another alternative is to find a function $C : \mathcal{E} \rightarrow [0, \infty)$ and $f : \mathbb{N} \rightarrow [0, \infty)$ such that

$$\forall \nu \in \mathcal{E}, \quad R_n(\pi, \nu) \leq C(\nu)f(n).$$

Bayesian regret is another candidate objective. For prior probability measure Q on $\mathcal{E}$, the Bayesian regret is the average of the regret with respect to the prior Q , i.e.,

$$\text{BR}_n(\pi, Q) = \int_{\mathcal{E}} R_n(\pi, \nu) dQ(\nu)$$

where the regret R_n is measurable.

Definition. Let $\nu = (P_a : a \in \mathcal{A})$ be a stochastic bandit. Then the **suboptimality gap** of action a is defined by

$$\Delta_a(\nu) := \mu^*(\nu) - \mu_a(\nu).$$

Also let

$$T_a(t) := \sum_{s=1}^t \mathbb{1}\{A_s = a\}$$

be the number of times action a has been chosen up to time t .

Lemma 4.5 (Regret decomposition lemma). *For any policy π and stochastic bandit ν with $\mathcal{A}$ finite or countable and horizon $n \in \mathbb{N}$, the regret R_n of policy π in ν satisfies*

$$R_n(\pi, \nu) = \sum_{a \in \mathcal{A}} \Delta_a(\nu) \mathbb{E}[T_a(n)].$$

Proof. For any $t \in \mathbb{N}$, $\sum_{a \in \mathcal{A}} \mathbb{1}\{A_t = a\} = 1$. Hence $S_n = \sum_t X_t = \sum_t \sum_a X_t \mathbb{1}\{A_t = a\}$. That is

$$R_n = n\mu^* - \mathbb{E}[S_n] = \sum_{a \in \mathcal{A}} \sum_{t=1}^n \mathbb{E}[(\mu^* - X_t) \mathbb{1}\{A_t = a\}].$$

Since $\mathbb{E}[X_t \mid H_{t-1}, A_t] = \mu_{A_t}$, we have

$$\begin{aligned} \mathbb{E}[(\mu^* - X_t) \mathbb{1}\{A_t = a\} \mid A_t] &= \mathbb{1}\{A_t = a\} \mathbb{E}[\mu^* - X_t \mid A_t] \\ &= \mathbb{1}\{A_t = a\} (\mu^* - \mu_a) \\ &= \mathbb{1}\{A_t = a\} \Delta_a. \end{aligned}$$

Therefore

$$\begin{aligned} R_n &= \sum_{a \in \mathcal{A}} \sum_{t=1}^n \mathbb{E}[(\mu^* - X_t) \mathbb{1}\{A_t = a\}] \\ &= \sum_{a \in \mathcal{A}} \sum_{t=1}^n \mathbb{E}[\mathbb{E}[(\mu^* - X_t) \mathbb{1}\{A_t = a\} \mid A_t]] \\ &= \sum_{a \in \mathcal{A}} \sum_{t=1}^n \mathbb{E}[\Delta_a \mathbb{1}\{A_t = a\}] \\ &= \sum_{a \in \mathcal{A}} \Delta_a \mathbb{E}[T_a(n)]. \end{aligned}$$

□

Exercise 4.10. Suppose ν is a finite-armed stochastic bandit and π is a policy such that

$$\lim_{n \rightarrow \infty} \frac{R_n(\pi, \nu)}{n} = 0.$$

Let $T^*(n) = \sum_{t=1}^n \mathbb{1}\{\mu_{A_t} = \mu^*\}$ be the number of times an optimal arm is chosen. Prove or disprove following statement:

- (a) $\lim_{n \rightarrow \infty} \mathbb{E}[T^*(n)]/n = 1$.
(b) $\lim_{n \rightarrow \infty} \mathbb{P}(\Delta_{A_t} > 0) = 0$.

Solution. (a) True, the following are the proof;

Proof. If every arm is optimal, then $\lim_{n \rightarrow \infty} \frac{\mathbb{E}[T^*(n)]}{n} = \lim_{n \rightarrow \infty} \frac{n}{n} = 1$. Suppose not. That is,

$$\exists a_0 \in \operatorname{argmin}_{a \in \{b \in \mathcal{A}: \mu_b < \mu^*\}} \{\mu^* - \mu_a\}.$$

Letting $c = \mu^* - \mu_{a_0}$ follows that $\mathbb{E}[(\mu^* - \mu_{A_t})] \geq c \cdot \mathbb{P}[\mu^* \neq \mu_{A_t}]$. Hence

$$\begin{aligned} \frac{R_n}{n} &= \sum_{t=1}^n \frac{\mathbb{E}[\mu^* - \mu_{A_t}]}{n} \\ &\geq c \frac{\sum_{t=1}^n \mathbb{P}[\mu^* \neq \mu_{A_t}]}{n}. \\ \therefore \lim_{n \rightarrow \infty} \frac{\sum_{t=1}^n \mathbb{P}[\mu^* \neq \mu_{A_t}]}{n} &= 0. \end{aligned}$$

Hence $\forall n \in \mathbb{N}$,

$$\begin{aligned} \frac{\mathbb{E}[T^*(n)]}{n} &= \sum_{t=1}^n \frac{\mathbb{E}[\mathbb{1}_{\mu_{A_t} = \mu^*}]}{n} \\ &= \frac{\sum_{t=1}^n \mathbb{P}[\mu_{A_t} = \mu^*]}{n} \\ &= 1 - \frac{\sum_{t=1}^n \mathbb{P}[\mu_{A_t} \neq \mu^*]}{n}. \\ \lim_{n \rightarrow \infty} \frac{\mathbb{E}[T^*(n)]}{n} &= 1 - \lim_{n \rightarrow \infty} \frac{\sum_{t=1}^n \mathbb{P}[\mu_{A_t} \neq \mu^*]}{n} = 1. \end{aligned}$$

□

- (b) False, the following is a counterexample;

Consider $\mathcal{A} = \{a_0, a_1\}$ with $\mu^* = \mu_{a_0}$ and $\mu^* - \mu_{a_1} = \Delta_{a_1} > 0$. Let the policy π choose a_1 at 2^k rounds for $k \in \mathbb{N}$. Then

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{R_n(\pi)}{n} &= \lim_{n \rightarrow \infty} \frac{\sum_{t=1}^n \mathbb{E}[\mu^* - \mu_{A_t}]}{n} \\ &\leq \lim_{n \rightarrow \infty} \frac{\sum_{t=1}^{\lfloor \log_2 n + 100 \rfloor} \Delta_{a_1}}{n} \\ &= \lim_{n \rightarrow \infty} \frac{\Delta_{a_1} \lfloor \log_2 n + 100 \rfloor}{n} \\ &= 0. \end{aligned}$$

But $\lim_{n \rightarrow \infty} \mathbb{P}[\Delta_{A_{2^k}} > 0] = 1$ for $k \in \mathbb{N}$, which disproves (b).

Exercise 4.11 (Retirement policy). Let $\mathcal{M}_1$ be a set of distributions on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, with finite means and $\mathcal{M}_2 = \{\delta_{\mu_2}\}$ be a singleton set containing the Dirac measure at μ_2 . Construct the set of bandits $\mathcal{E} = \mathcal{M}_1 \times \mathcal{M}_2$. Here a policy $\pi = (\pi_t)_{t \in \mathbb{N}}$ is called a retirement policy if $a_t = 2$, then

$$\pi_{t+1}(2 \mid a_1, x_1, \dots, a_t, x_t) = 1, \quad \forall (a_s)_{s=1}^{t-1} \text{ and } (x_s)_{s=1}^t.$$

- (a) Let n be fixed and $\pi = (\pi_t)_{t=1}^n$ be any policy. Prove there exists a retirement policy $\pi' = (\pi'_t)_{t=1}^n$ such that for all $\nu \in \mathcal{E}$,

$$R_n(\pi', \nu) \leq R_n(\pi, \nu)$$

- (b) Let $\mathcal{M}_1 = \{\mathcal{B}(\mu_1) : \mu_1 \in [0, 1]\}$ and suppose that $\pi = (\pi_t)_{t=1}^\infty$ is a retirement policy. Prove there exists a bandit $\nu \in \mathcal{E}$ such that

$$\limsup_{n \rightarrow \infty} \frac{R_n(\pi, \nu)}{n} > 0.$$

Proof. (a) Let

$$\mathcal{H}_j := (\{1, 2\} \times \mathbb{R})^j, \quad j \in \{0, \dots, n\},$$

where $\mathcal{H}_0 = \{\emptyset\}$. Fix $n \in \mathbb{N}$ and a policy $\pi = (\pi_j)_{j=1}^n$. Let $U_1, \dots, U_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, 1)$. For $j \in \{0, \dots, n-1\}$ and $h \in \mathcal{H}_j$, define

$$\psi_{j+1}(h) := \begin{cases} 1, & U_{j+1} \leq \pi_{j+1}(1 | h), \\ 2, & U_{j+1} > \pi_{j+1}(1 | h). \end{cases}$$

Define

$$\Gamma : \bigcup_{j=0}^n \mathcal{H}_j \longrightarrow \bigcup_{j=0}^n \mathcal{H}_j$$

such that $\forall j \in \{0, \dots, n\}$ and $\forall h \in \mathcal{H}_j$,

$$\Gamma(h) := \begin{cases} h, & j = n, \\ h, & j < n \text{ and } \psi_{j+1}(h) = 1, \\ \Gamma(h, (2, \mu_2)), & j < n \text{ and } \psi_{j+1}(h) = 2. \end{cases}$$

Now fix $\nu = (\nu_1, \delta_{\mu_2}) \in \mathcal{E}$. Let

$$Y_1, Y_2, \dots \stackrel{\text{i.i.d.}}{\sim} \nu_1$$

be the rewards obtained from arm 1. Define $\tilde{H}_0, \dots, \tilde{H}_n$ and $A'_1, \dots, A'_n$ by

$$\tilde{H}_0 := \Gamma(\emptyset),$$

and, for $t \in \{1, \dots, n\}$,

$$A'_t := \begin{cases} 1, & |\tilde{H}_{t-1}| < n, \\ 2, & |\tilde{H}_{t-1}| = n, \end{cases}$$

and

$$\tilde{H}_t := \begin{cases} \Gamma(\tilde{H}_{t-1}, (1, Y_t)), & |\tilde{H}_{t-1}| < n, \\ \tilde{H}_{t-1}, & |\tilde{H}_{t-1}| = n. \end{cases}$$

Then $\forall t \in \{0, \dots, n\}$, $\tilde{H}_t \in \bigcup_{j=0}^n \mathcal{H}_j$. Define $\forall h \in \mathcal{H}_{t-1}$ and $\forall a \in \{1, 2\}$,

$$H'_{t-1} \in \mathcal{H}_{t-1} \text{ and } \pi'_t(a | h) = \mathbb{P}(A'_t = a | H'_{t-1} = h).$$

for all $t \in \{1, \dots, n\}$. We first show that π' is a retirement policy. Suppose that $A'_{t_0} = 2$ for some $t_0 \in \{1, \dots, n\}$. Then

$$|\tilde{H}_{t_0-1}| = n.$$

Hence, by the definition of $\tilde{H}_t$,

$$\tilde{H}_s = \tilde{H}_{t_0-1} \quad \text{for all } s \geq t_0 - 1.$$

Therefore,

$$A'_s = 2 \quad \text{for all } s \geq t_0.$$

Thus, π' is a retirement policy. Let

$$K := T_1^{\pi'}(n) = \sum_{t=1}^n \mathbb{1}_{\{A'_t=1\}}.$$

By construction,

$$\tilde{H}_K \in \mathcal{H}_n.$$

Moreover, conditionally on each prefix of $\tilde{H}_K$, its next action is chosen according to the same conditional distribution as under π , and its next reward has distribution ν_1 if that action is 1, and equals μ_2 if that action is 2. Consequently,

$$\tilde{H}_K \stackrel{d}{=} H_n^\pi,$$

where H_n^π denotes the n -round history generated by π in the bandit ν . Since K is exactly the number of occurrences of action 1 in $\tilde{H}_K$, it follows that

$$T_1^{\pi'}(n) \stackrel{d}{=} T_1^\pi(n).$$

Hence,

$$\mathbb{E}_\nu^{\pi'}[T_1^{\pi'}(n)] = \mathbb{E}_\nu^\pi[T_1^\pi(n)].$$

Since $T_2(n) = n - T_1(n)$,

$$\mathbb{E}_\nu^{\pi'}[T_2^{\pi'}(n)] = \mathbb{E}_\nu^\pi[T_2^\pi(n)].$$

Finally, by the regret decomposition lemma,

$$\begin{aligned} R_n(\pi', \nu) &= \Delta_1(\nu) \mathbb{E}_\nu^{\pi'}[T_1^{\pi'}(n)] + \Delta_2(\nu) \mathbb{E}_\nu^{\pi'}[T_2^{\pi'}(n)] \\ &= \Delta_1(\nu) \mathbb{E}_\nu^\pi[T_1^\pi(n)] + \Delta_2(\nu) \mathbb{E}_\nu^\pi[T_2^\pi(n)] \\ &= R_n(\pi, \nu). \end{aligned}$$

Therefore,

$$R_n(\pi', \nu) \leq R_n(\pi, \nu)$$

for every $\nu \in \mathcal{E}$.

(b) If $\mu_2 \geq 1$ or $\mu_2 \leq 0$, then (b) is false (Consider trivial policy, choosing optimal arm all the time). Let's suppose $\mu_2 \in (0, 1)$. Then

$$\exists p, q \in (0, 1) \text{ such that } 0 < p < \mu_2 < q < 1.$$

Define $\nu_p = (\mathcal{B}(p), \delta_{\mu_2})$, $\nu_q = (\mathcal{B}(q), \delta_{\mu_2})$ and $\tau = \inf\{t \in \mathbb{N} : A_t = 2\}$, also let a retirement policy π be given. Then $T_1(n) = \min\{\tau - 1, n\}$, and $T_2(n) = n - T_1(n)$. If

$$\limsup_{n \rightarrow \infty} \frac{R_n(\pi, \nu_p)}{n} > 0,$$

then the conclusion follows. Suppose

$$\limsup_{n \rightarrow \infty} \frac{R_n(\pi, \nu_p)}{n} = 0.$$

Then

$$\limsup_{n \rightarrow \infty} \frac{(\mu_2 - p) \mathbb{E}_{\nu_p, \pi}[T_1(n)]}{n} = 0.$$

But $T_1(n)/n = \min\{\tau - 1, n\}/n \rightarrow \mathbb{1}_{\{\tau = \infty\}}$ almost surely. Since $0 \leq T_1(n)/n \leq 1$, by the dominated convergence theorem,

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}_{\nu_p, \pi}[T_1(n)]}{n} = \mathbb{P}_{\nu_p, \pi}[\tau = \infty] = 0.$$

Therefore,

$$\begin{aligned} \mathbb{P}_{\nu_q, \pi}[\tau = \infty] &= 1 \\ \implies \mathbb{P}_{\nu_q, \pi} \left[\bigcup_{m=1}^{\infty} \{\tau \leq m\} \right] &= 1 \\ \implies \exists m \in \mathbb{N} \text{ such that } \mathbb{P}_{\nu_q, \pi}[\tau \leq m] &> 0. \end{aligned}$$

Here $\mathbb{P}_{\nu_p} < \mathbb{P}_{\nu_q}$ since $\forall y = (y_1, y_2, \dots) \in \{0, 1\}^m$,

$$\begin{aligned}\mathbb{P}_{\nu_p}[Y_1 = y_1, \dots, Y_m = y_m] &= p^{\sum_{i=1}^m y_i} (1-p)^{m - \sum_{i=1}^m y_i} > 0 \\ \implies \mathbb{P}_{\nu_q}[Y_1 = y_1, \dots, Y_m = y_m] &= q^{\sum_{i=1}^m y_i} (1-q)^{m - \sum_{i=1}^m y_i} > 0.\end{aligned}$$

Hence if $\mathbb{P}_{\nu_q}[\tau \leq m] > 0$, then $\mathbb{P}_{\nu_p}[\tau \leq m] > 0$. Now let $c = \mathbb{P}_{\nu_q}[\tau \leq m] > 0$. Then we have

$$\Delta_2(\nu_q) = q - \mu_2 > 0, \quad \text{and} \quad T_2(n) \geq (n - m + 1) \mathbb{1}_{\{\tau \leq m\}}.$$

Therefore

$$\begin{aligned}\limsup_{n \rightarrow \infty} \frac{R_n(\pi, \nu_q)}{n} &\geq \liminf_{n \rightarrow \infty} \frac{R_n(\pi, \nu_q)}{n} \\ &\geq \liminf_{n \rightarrow \infty} (q - \mu_2) \frac{\mathbb{E}_{\nu_q, \pi}[T_2(n)]}{n} \\ &\geq \liminf_{n \rightarrow \infty} (q - \mu_2) \frac{\mathbb{E}_{\nu_q, \pi}[(n - m + 1) \mathbb{1}_{\{\tau \leq m\}}]}{n} \\ &= (q - \mu_2) \left(1 - \frac{m - 1}{n}\right) \mathbb{P}_{\nu_q, \pi}[\tau \leq m] \\ &= (q - \mu_2) c \\ &> 0.\end{aligned}$$

□

1.5 Concentration of measure

Definition. Let $X, X_1, X_2, \dots, X_n$ be i.i.d. random variables with real values, and assume that the mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \mathbb{V}[X] < \infty$. Then the **empirical mean** is defined by

$$\hat{\mu} := \frac{1}{n} \sum_{i=1}^n X_i.$$

This is an unbiased estimator of the mean μ , since $\mathbb{E}[\hat{\mu}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mu$. Using independence of the X_i 's, the variance of the empirical mean is

$$\mathbb{V}[\hat{\mu}] = \mathbb{E}[(\hat{\mu} - \mu)^2] = \frac{\sigma^2}{n}.$$

Two quantities

$$\mathbb{P}[\hat{\mu} \geq \mu + \epsilon] \quad \text{and} \quad \mathbb{P}[\hat{\mu} \leq \mu - \epsilon]$$

are called the **tail probabilities** of $\hat{\mu} - \mu$.

Lemma 5.1. *Let X be a random variable and let $\epsilon > 0$. Then*

$$(a) \text{ (Markov's inequality) : } \mathbb{P}[|X| \geq \epsilon] \leq \frac{\mathbb{E}[|X|]}{\epsilon}.$$

$$(b) \text{ (Chebyshev's inequality) : } \mathbb{P}[|X - \mu| \geq \epsilon] \leq \frac{\sigma^2}{\epsilon^2}.$$

Proof. (a)

$$\begin{aligned}\mathbb{E}[|X|] &= \int_0^\infty \mathbb{P}[|X| \geq t] dt \\ &\geq \int_0^\epsilon \mathbb{P}[|X| \geq \epsilon] dt \\ &= \epsilon \mathbb{P}[|X| \geq \epsilon].\end{aligned}$$

(b)

$$\begin{aligned}\mathbb{P}[|X - \mu| \geq \epsilon] &= \mathbb{P}[(X - \mu)^2 \geq \epsilon^2] \\ &\leq \frac{\mathbb{E}[(X - \mu)^2]}{\epsilon^2}.\end{aligned}$$

□

Theorem (Central Limit Theorem). *Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with finite second moment. Define*

$$S_n := \sum_{t=1}^n (X_t - \mu).$$

Then for $\sigma^2 > 0$,

$$\frac{S_n}{\sqrt{n}\sigma} \xrightarrow{d} N(0, 1).$$

Definition 5.2. A random variable X is σ -subgaussian if

$$\forall \lambda \in \mathbb{R}, \quad \mathbb{E}[e^{\lambda X}] \leq e^{\frac{\lambda^2 \sigma^2}{2}}.$$

The **moment generating function** of X is a function

$$M_X(\lambda) := \mathbb{E}[e^{\lambda X}], \quad \lambda \in \mathbb{R}.$$

The **cumulant generating function** of X is a function

$$\Lambda_X(\lambda) := \log M_X(\lambda) = \log \mathbb{E}[e^{\lambda X}], \quad \lambda \in \mathbb{R}.$$

Remark. X is σ -subgaussian if and only if

$$\Lambda_X(\lambda) \leq \frac{\lambda^2 \sigma^2}{2} \quad \forall \lambda \in \mathbb{R}.$$

Theorem 5.3. *If X is σ -subgaussian, then for all $\epsilon \geq 0$,*

$$\mathbb{P}[X \geq \epsilon] \leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right).$$

Proof. Let $\lambda > 0$. By Markov's inequality,

$$\mathbb{P}[X \geq \epsilon] = \mathbb{P}[e^{\lambda X} \geq e^{\lambda \epsilon}] \leq \frac{\mathbb{E}[e^{\lambda X}]}{e^{\lambda \epsilon}} \leq \frac{e^{\frac{\lambda^2 \sigma^2}{2}}}{e^{\lambda \epsilon}} = \exp\left(\frac{\lambda^2 \sigma^2}{2} - \lambda \epsilon\right).$$

Letting $\lambda = \frac{\epsilon}{\sigma^2}$ gives

$$\mathbb{P}[X \geq \epsilon] \leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right).$$

□

Lemma 5.4. *Suppose X is σ -subgaussian and X_1, X_2 are independent σ_1, σ_2 -subgaussian, respectively. Then*

- (a) $\mathbb{E}[X] = 0$ and $\mathbb{V}[X] \leq \sigma^2$.
- (b) cX is $|c|\sigma$ -subgaussian for all $c \in \mathbb{R}$.
- (c) $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$ -subgaussian.

Proof. $\log x$ is convex on $(0, \infty)$, in the sense that for any $x, y > 0$ and $\theta \in [0, 1]$,

$$\log(\theta x + (1 - \theta)y) \geq \theta \log(x) + (1 - \theta) \log(y).$$

- (a) By Jensen's inequality, $\log e^{\frac{\lambda^2 \sigma^2}{2}} \geq \log \mathbb{E}[e^{\lambda X}] \geq \mathbb{E}[\log e^{\lambda X}] = \lambda \mathbb{E}[X]$. That is, $\mathbb{E}[X] \leq \frac{\lambda \sigma^2}{2}$ for all $\lambda \in \mathbb{R}$, which implies $\mathbb{E}[X] = 0$. Also,

$$\mathbb{E}[e^{\lambda X}] + \mathbb{E}[e^{-\lambda X}] \leq 2e^{\frac{\lambda^2 \sigma^2}{2}}, \quad \forall \lambda \in \mathbb{R}.$$

Hence,

$$\mathbb{E}\left[\frac{e^{\lambda X} - e^{-\lambda X} - 2}{\lambda^2}\right] \leq \frac{e^{\frac{2(\lambda^2 \sigma^2)}{2} - 1}}{\lambda^2}.$$

Here

$$\frac{e^{\lambda X} - e^{-\lambda X}}{\lambda^2} = \frac{2(\cosh(\lambda X) - 1)}{\lambda^2},$$

which is non-negative and $\frac{2(\cosh(\lambda X) - 1)}{\lambda^2} \rightarrow X^2$ as $\lambda \rightarrow 0$. Therefore, by Fatou's lemma,

$$\mathbb{E}[X^2] \leq \liminf_{\lambda \rightarrow 0} \mathbb{E}\left[\frac{2(\cosh(\lambda X) - 1)}{\lambda^2}\right] \leq \lim_{\lambda \rightarrow 0} \frac{e^{\frac{2(\lambda^2 \sigma^2)}{2} - 1}}{\lambda^2} = \sigma^2.$$

- (b) $\forall \lambda \in \mathbb{R}$, $\mathbb{E}[e^{\lambda cX}] = \mathbb{E}[e^{(c\lambda)X}] \leq e^{\frac{(c\lambda)^2 \sigma^2}{2}} = e^{\frac{\lambda^2 (c\sigma)^2}{2}}$, which shows that cX is $|c|\sigma$ -subgaussian.
- (c) $\forall \lambda \in \mathbb{R}$, $\mathbb{E}[e^{\lambda(X_1 + X_2)}] = \mathbb{E}[e^{\lambda X_1}] \mathbb{E}[e^{\lambda X_2}] \leq e^{\frac{\lambda^2 \sigma_1^2}{2}} e^{\frac{\lambda^2 \sigma_2^2}{2}} = e^{\frac{\lambda^2 (\sigma_1^2 + \sigma_2^2)}{2}}$, which shows that $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$ -subgaussian. □

Corollary 5.5. Assume $X_i - \mu$ are independent, σ -subgaussian random variables. Then for any $\epsilon \geq 0$,

$$\mathbb{P}(\hat{\mu} \geq \mu + \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right) \quad \text{and} \quad \mathbb{P}(\hat{\mu} \leq \mu - \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right),$$

where $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$.

Proof. By Lemma 5.4, $\hat{\mu} - \mu$ is $\frac{\sigma}{\sqrt{n}}$ -subgaussian. Therefore, for any $\epsilon \geq 0$,

$$\mathbb{P}(\hat{\mu} \geq \mu + \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right) \quad \text{and} \quad \mathbb{P}(\hat{\mu} \leq \mu - \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right)$$

by Theorem 5.3. □

Exercise 5.6.

Exercise 5.9.

Exercise 5.10.

1.6 The Explore-Then-Commit Algorithm

From now on, we assume that all bandit instances are in $\mathcal{E}_{\text{SG}}^k(1)$, which means the reward distribution of all arms is 1-subgaussian. Let $\hat{\mu}_i(t)$ be the average reward received from arm i up to round t . Formally,

$$\hat{\mu}_i(t) = \frac{1}{T_i(t)} \sum_{s=1}^t \mathbb{1}\{A_s = i\} X_s,$$

where $T_i(t) = \sum_{s=1}^t \mathbb{1}\{A_s = i\}$ is the number of times arm i has been played up to round t .

Input: m

1 In round t choose action

$$A_t = \begin{cases} (t \bmod k) + 1, & \text{if } t \leq mk, \\ \operatorname{argmax}_i \hat{\mu}_i(mk) & \text{if } t > mk, \end{cases}$$

Algorithm 1: ETC Algorithm

Theorem 6.1. Suppose ETC with 1-subgaussian bandit and $1 \leq m \leq n/k$. Then

$$R_n \leq m \sum_{i=1}^k \Delta_i + (n - mk) \sum_{i=1}^k \Delta_i \exp\left(-\frac{m\Delta_i^2}{4}\right)$$

Proof. Fill. □

Exercise 6.6 (Doubling trick). Let a fixed sequence $\{n_i\}_{i=1}^\infty$ be given and the policy π be given with following regret bound;

$$\max_{1 \leq t \leq n} R_t(\pi(n), \nu) \leq f_n(\nu), \quad \forall \nu \in \mathcal{E}.$$

Consider the policy π^* that runs π until round $t = \min\{n, n_1\}$, then restarts π until round $t = \min\{n, n_1 + n_2\}$, and so on.

(a) Let $n > 0$ be arbitrary, $l_{\max} = \min\{l : \sum_{i=1}^l n_i \geq n\}$. Prove that $\forall \nu \in \mathcal{E}$,

$$R_n(\pi^*, \nu) \leq \sum_{l=1}^{l_{\max}} f_{n_l}(\nu)$$

(b) Suppose that $f_n(\nu) \leq \sqrt{n}$. Prove that if $n_l = 2^{l-1}$, then for any $\nu \in \mathcal{E}$ and horizon n ,

$$R_n(\pi^*, \nu) \leq C\sqrt{n}.$$

(c) Suppose that $f_n(\nu) = g(\nu) \log(n)$ for some function $g : \mathcal{E} \rightarrow [0, \infty)$. What is the regret of π^* if $n_l = 2^{l-1}$?

Solution .

Exercise 6.8.

1.7 The Upper Confidence Bound Algorithm

Note that for sequence of independent 1-subgaussian random variables $(X_t)_{t=1}^n$, mean μ and the empirical mean $\hat{\mu}_n = \frac{1}{n} \sum_{t=1}^n X_t$ satisfy

$$\mathbb{P}\left(\mu \geq \hat{\mu}_n + \sqrt{\frac{2 \log(1/\delta)}{n}}\right) \leq \delta, \quad \forall \delta \in (0, 1).$$

Define

$$\text{UCB}_i(t-1, \delta) = \begin{cases} \infty, & \text{if } T_i(t-1) = 0, \\ \hat{\mu}_i(t-1) + \sqrt{\frac{2 \log(1/\delta)}{T_i(t-1)}}, & \text{if } T_i(t-1) > 0. \end{cases}$$

```

1 Input  $k$  and  $\delta$ 
2 for  $t \in \{1, \dots, n\}$  do
3   Choose action

```

$$A_t = \underset{i}{\operatorname{argmax}} \text{UCB}_i(t-1, \delta)$$

```

4   Observe reward  $X_t$  and update upper confidence bounds
5 end for

```

Algorithm 2: UCB(δ)

Theorem 7.1. *Suppose UCB with stochastic k -armed 1-subgaussian bandits. Then for all n , if $\delta = 1/n$, then*

$$R_n \leq 3 \sum_{i=1}^k \Delta_i + \sum_{i: \Delta_i > 0} \frac{16 \log(n)}{\Delta_i}$$

Proof. WLOG, assume that $\mu_1 = \mu^*$. Let G_i be the 'good' event defined by

$$G_i = \left\{ \mu_1 < \min_{t \in \{1, \dots, n\}} \text{UCB}_1(t, \delta) \right\} \cap \left\{ \hat{\mu}_{i, u_i} + \sqrt{\frac{2}{u_i} \log \left(\frac{1}{\delta} \right)} < \mu_1 \right\}$$

where $u_i \in \{1, \dots, n\}$. Assume that G_i holds for all $i \in \{1, \dots, k\}$. To make contradiction, suppose that $T_i(n) > u_i$. Then there exists $t \in \{1, \dots, n\}$ such that $T_i(t-1) = u_i$ and $A_t = i$. Hence

$$\begin{aligned} \text{UCB}_i(t-1, \delta) &= \hat{\mu}_i(t-1) + \sqrt{\frac{2 \log(1/\delta)}{T_i(t-1)}} \\ &= \hat{\mu}_{i, u_i} + \sqrt{\frac{2}{u_i} \log \left(\frac{1}{\delta} \right)} \\ &< \mu_1 \\ &\leq \text{UCB}_1(t-1, \delta). \end{aligned}$$

Therefore, $A_t = \arg\max_j \text{UCB}_j(t-1, \delta) \neq i$, which is a contradiction. On the other hand,

$$G_i^c = \left\{ \mu_1 \geq \min_{t \in \{1, \dots, n\}} \text{UCB}_1(t, \delta) \right\} \cup \left\{ \hat{\mu}_{i, u_i} + \sqrt{\frac{2}{u_i} \log \left(\frac{1}{\delta} \right)} \geq \mu_1 \right\}.$$

Former part bounded by

$$\begin{aligned} \mathbb{P} \left(\mu_1 \geq \min_{t \in \{1, \dots, n\}} \text{UCB}_1(t, \delta) \right) &\leq \mathbb{P} \left(\mu_1 \geq \min_{s \in \{1, \dots, n\}} \hat{\mu}_1(s) + \sqrt{\frac{2}{s} \log \left(\frac{1}{\delta} \right)} \right) \\ &\leq \mathbb{P} \left(\bigcup_{s \in \{1, \dots, n\}} \left\{ \mu_1 \geq \hat{\mu}_1(s) + \sqrt{\frac{2}{s} \log \left(\frac{1}{\delta} \right)} \right\} \right) \\ &\leq \sum_{s \in \{1, \dots, n\}} \mathbb{P} \left(\mu_1 \geq \hat{\mu}_1(s) + \sqrt{\frac{2}{s} \log \left(\frac{1}{\delta} \right)} \right) \\ &\leq \sum_{s \in \{1, \dots, n\}} \delta = n\delta. \end{aligned}$$

For latter part, assume u_i is chosen large enough that

$$\Delta_i - \sqrt{\frac{2}{u_i} \log \left(\frac{1}{\delta} \right)} \geq c\Delta_i$$

for some $c \in (0, 1)$. Then

$$\begin{aligned} \mathbb{P} \left(\hat{\mu}_{i, u_i} + \sqrt{\frac{2}{u_i} \log \left(\frac{1}{\delta} \right)} \geq \mu_1 \right) &= \mathbb{P} \left(\hat{\mu}_{i, u_i} - \mu_i \geq \Delta_i - \sqrt{\frac{2}{u_i} \log \left(\frac{1}{\delta} \right)} \right) \\ &\leq \mathbb{P} (\hat{\mu}_{i, u_i} - \mu_i \geq c\Delta_i) \\ &\leq \exp \left(-\frac{u_i (c\Delta_i)^2}{2} \right) \end{aligned}$$

by Cor.5.5. Now we have

$$\mathbb{P}(G_i^c) \leq n\delta + \exp\left(-\frac{u_i(c\Delta_i)^2}{2}\right).$$

Also

$$\mathbb{E}[T_i(n)] = \mathbb{E}[\mathbb{1}_{G_i} T_i(n)] + \mathbb{E}[\mathbb{1}_{G_i^c} T_i(n)].$$

Therefore, choosing

$$u_i = \left\lceil \frac{2 \log(1/\delta)}{((1-c)\Delta_i)^2} \right\rceil$$

follows that

$$\mathbb{E}[T_i(n)] \leq u_i + 1 + n^{1-2c^2/(1-c)^2} = \left\lceil \frac{2 \log(n^2)}{((1-c)\Delta_i)^2} \right\rceil + 1 + n^{1-2c^2/(1-c)^2}.$$

Lastly, choosing $c = 1/2$ follows that

$$\mathbb{E}[T_i(n)] \leq \left\lceil \frac{16 \log(n)}{\Delta_i^2} \right\rceil + 2,$$

and regret decomposition lemma leads to

$$\begin{aligned} R(n) &\leq \sum_{i:\Delta_i > 0} \Delta_i \left(\left\lceil \frac{16 \log(n)}{\Delta_i^2} \right\rceil + 2 \right) \\ &= \sum_{i:\Delta_i > 0} \left(\frac{16 \log(n)}{\Delta_i} + \Delta_i \right) + 2 \sum_{i:\Delta_i > 0} \Delta_i \\ &\leq 3 \sum_{i=1}^k \Delta_i + \sum_{i:\Delta_i > 0} \frac{16 \log(n)}{\Delta_i}. \end{aligned}$$

□

Theorem 7.2. Suppose $\nu \in \mathcal{E}_{\text{SG}}^k(1)$ and $\delta = 1/n^2$. Then regret of the UCB algorithm is bounded by

$$R(n) \leq 8\sqrt{nk \log(n)} + 3 \sum_{i=1}^k \Delta_i.$$

Proof. Let $\Delta > 0$ be some value to be determined later. Recall from proof of 7.1,

$$\mathbb{E}[T_i(n)] \leq 3 + \frac{16 \log(n)}{\Delta_i^2}.$$

Hence regret decomposition lemma follows that

$$\begin{aligned} R_n &= \sum_{i:\Delta_i > 0} \Delta_i \mathbb{E}[T_i(n)] \\ &= \sum_{i:\Delta_i < \Delta} \Delta_i \mathbb{E}[T_i(n)] + \sum_{i:\Delta_i \geq \Delta} \Delta_i \mathbb{E}[T_i(n)] \\ &\leq n\Delta + \sum_{i:\Delta_i \geq \Delta} \left(3\Delta_i + \frac{16 \log(n)}{\Delta_i} \right) \\ &\leq n\Delta + 3 \sum_{i=1}^k \Delta_i + \frac{16k \log(n)}{\Delta} \\ &\leq 8\sqrt{nk \log(n)} + 3 \sum_{i=1}^k \Delta_i. \end{aligned}$$

by choosing $\Delta = \sqrt{\frac{16k \log(n)}{n}}$.

□

```

1 Input  $k$  and  $\delta$ 
2 Choose each arm once
3 for  $l = 1, 2, \dots$  do
4   Compute  $A_l = \operatorname{argmax}_i \text{UCB}_i(t-1, \delta)$ 
5   Choose arm  $A_l$  and exactly  $2^l$  times
6 end for

```

Algorithm 3: A phased version of UCB

Exercise 7.4 (Phased UCB 1). Consider the following algorithm, called phased UCB;

- (a) State and prove a bound on the regret for this version of UCB.
- (b) Compare with Theorem 7.1.
- (c) How would the result change if the l th phase had a length of $\lceil \alpha^l \rceil$ with $\alpha > 1$?

Solution.

- (a) Let $k = 2$, L be some large constant (≥ 3), $n = 2^{L+1}$, $\delta = n^{-2}$, and $q = 2\sqrt{\log(n)}$. Also let two arms attain deterministic reward, $\mu_1 = 0$, $\mu_2 = -\Delta$, respectively. ($\Delta > 0$) In this setting, the algorithm chooses each arm once initially and then goes to round 3. Here learner gets UCB

$$\text{UCB}_1(2, \delta) = q, \quad \text{UCB}_2(2, \delta) = -\Delta + q$$

and hence chooses arm 1 in round 3, and 4. Explicit simulation will be following;

t	l	UCB_1	UCB_2	Next action
1	0	∞	∞	1
2	0	q	∞	2
3	1	q	$-\Delta + q$	1
4	1	$q/\sqrt{2}$	$-\Delta + q$	1
5	2	$q/\sqrt{3}$	$-\Delta + q$	1
...	...	...	...	...
2^L	$L-1$	$q/\sqrt{2^L-2}$	$-\Delta + q$	1
$2^L + 1$	L	$q/\sqrt{2^L-1}$	$-\Delta + q$	2
...	...	...	...	...

if we choose Δ with

$$\frac{q}{\sqrt{2^L-1}} < q - \Delta < \frac{q}{\sqrt{2^{L-1}-1}}.$$

Then the regret will be dominated by the suboptimal arm 2 in the last phase, giving

$$R_n \geq \Delta \cdot 2^L = \frac{n}{2} \Delta.$$

This example shows that the phased UCB algorithm can obtain linear regret,

$$R_n \leq n \cdot \max_i \Delta_i,$$

which is trivial.

- (b) The above example of (a) shows that the phased UCB algorithm can get linear regret, while Theorem 7.1 shows that the standard UCB always get lower than sublinear regret.
- (c) Still get linear regret for the phased UCB algorithm with any constant $\alpha > 1$.

Exercise 7.5 (Phased UCB 2). Consider the following another phased version of UCB;

```

1 Input  $k$  and  $\delta$ 
2 Choose each arm once
3 for  $l = 1, 2, \dots$  do
4   Let  $t_l = t$ 
5   Compute  $A_l = \operatorname{argmax}_i \operatorname{UCB}_i(t_l - 1, \delta)$ 
6   Choose arm  $A_l$  until round  $t$  such that  $T_{A_l}(t) \geq \alpha T_{A_l}(t_l - 1)$ 
7 end for

```

- (a) State and prove a bound on the regret for this version of UCB with $\alpha = 2$.
- (b) Compare with previous exercise and Theorem 7.1.
- (c) Repeat (a) and (b) for $\alpha > 1$.
- (d) Implement these algorithms and compare them empirically to $\operatorname{UCB}(\delta)$

Solution. WLOG, assume that arm 1 is optimal.

- (a) For all suboptimal arms i , let

$$H_i = \left\{ \forall t \in \{1, \dots, n\}, \mu_1 < \hat{\mu}_{1,t} + \sqrt{\frac{2 \log(1/\delta)}{t}} \right\} \cap \left\{ \forall t \in \{1, \dots, n\}, \mu_i > \hat{\mu}_{i,t} - \sqrt{\frac{2 \log(1/\delta)}{t}} \right\}.$$

Then the complement event H_i^c is

$$H_i^c = \left\{ \exists t \in \{1, \dots, n\} : \mu_1 \geq \hat{\mu}_{1,t} + \sqrt{\frac{2 \log(1/\delta)}{t}} \right\} \cup \left\{ \exists t \in \{1, \dots, n\} : \mu_i \leq \hat{\mu}_{i,t} - \sqrt{\frac{2 \log(1/\delta)}{t}} \right\}.$$

That is

$$\mathbb{P}(H_i^c) \leq \sum_{t=1}^n \mathbb{P} \left(\mu_1 \geq \hat{\mu}_{1,t} + \sqrt{\frac{2 \log(1/\delta)}{t}} \right) + \sum_{t=1}^n \mathbb{P} \left(\mu_i \leq \hat{\mu}_{i,t} - \sqrt{\frac{2 \log(1/\delta)}{t}} \right).$$

Cor 5.5 follows that

$$\mathbb{P}(H_i^c) \leq 2 \sum_{t=1}^n \exp \left(-\frac{2 \log(1/\delta)}{t} \cdot \frac{t}{2} \right) = 2 \sum_{t=1}^n \delta = 2n\delta.$$

For the latest phase l where arm i is chosen, let $m := T_i(t_l - 1)$. Upon the event H_i , letting $m \geq u_i$ implies that

$$\operatorname{UCB}_i(t_l - 1, \delta) = \hat{\mu}_{i,m} + \sqrt{\frac{2 \log(1/\delta)}{m}} < \mu_i + \sqrt{\frac{8 \log(1/\delta)}{m}} < \mu_i + \Delta_i = \mu_1 < \operatorname{UCB}_1(t_l - 1, \delta)$$

where $u_i = \lfloor \frac{8 \log(1/\delta)}{\Delta_i^2} \rfloor + 1$.

On the other hand, if $m < u_i$, then $T_i(n) \leq 2m < 2u_i$. Therefore

$$\begin{aligned} \mathbb{E}[T_i(n)] &\leq 2u_i + n\mathbb{P}(H_i^c) \\ &= 2u_i + 2n^2\delta. \end{aligned}$$

Choosing $\delta = n^{-2}$ follows that $u_i \leq \frac{16 \log(n)}{\Delta_i^2} + 1$, and hence

$$\mathbb{E}[T_i(n)] \leq 2 \left(\frac{16 \log(n)}{\Delta_i^2} + 1 \right) + 2 = \frac{32 \log(n)}{\Delta_i^2} + 4.$$

The regret is bounded by

$$\begin{aligned}
R(n) &= \sum_{i:\Delta_i>0} \Delta_i \mathbb{E}[T_i(n)] \\
&\leq \sum_{i:\Delta_i>0} \Delta_i \left(\frac{32 \log(n)}{\Delta_i^2} + 4 \right) \\
&= 4 \sum_{i=1}^k \Delta_i + 32 \sum_{i:\Delta_i>0} \frac{\log(n)}{\Delta_i}.
\end{aligned}$$

(b) Slightly larger than UCB, they differ by scalar factor.

(c) $T_i(n) \leq \lceil \alpha m \rceil$,

$$R(n) \lesssim 4 \sum_{i=1}^k \Delta_i + 16\alpha \sum_{i:\Delta_i>0} \frac{\log(n)}{\Delta_i}.$$

(d) Refer "./Implementations".

1.8 The Upper Confidence Bound Algorithm: Asymptotic Optimality

1 **Input** k
2 Choose each arm once
3 Subsequently choose

$$A_t = \operatorname{argmax}_i \left(\hat{\mu}_i(t-1) + \sqrt{\frac{2 \log f(t)}{T_i(t-1)}} \right)$$

where $f(t) = 1 + t \log^2(t)$.

Algorithm 4: Asymptotically optimal UCB

Theorem 8.1. *For any 1-subgaussian bandit, regret of the asymptotically optimal UCB policy satisfies*

$$R_n \leq \sum_{i:\Delta_i>0} \inf_{\epsilon \in (0, \Delta_i)} \Delta_i \left(1 + \frac{5}{\epsilon^2} + \frac{2 \left(\log f(n) + \sqrt{\pi \log f(n)} + 1 \right)}{(\Delta_i - \epsilon)^2} \right).$$

Furthermore,

$$\limsup_{n \rightarrow \infty} \frac{R_n}{\log n} \leq \sum_{i:\Delta_i>0} \frac{2}{\Delta_i}.$$

Choosing $\epsilon = \Delta_i/2$ inside the sum shows that

$$R_n \leq \sum_{i:\Delta_i>0} \left(\Delta_i + \frac{1}{\Delta_i} \left(8 \log f(n) + 8 \sqrt{\pi \log f(n)} + 28 \right) \right).$$

That is, $\exists C > 0$ such that

$$R_n \leq \sum_{i:\Delta_i>0} C \left(\Delta_i + \frac{\log n}{\Delta_i} \right).$$

Lemma 8.2. *Let $X_1, \dots, X_n$ be independent 1-subgaussian random variables, $\hat{\mu}_t = \frac{1}{t} \sum_{i=1}^t X_i$, $\epsilon > 0$, $a > 0$. Suppose that*

$$\kappa = \sum_{t=1}^n \mathbb{1}_{\{\hat{\mu}_t + \sqrt{\frac{2a}{t}} \geq \epsilon\}}, \quad \kappa' = u + \sum_{t=\lceil u \rceil}^n \mathbb{1}_{\{\hat{\mu}_t + \sqrt{\frac{2a}{t}} \geq \epsilon\}}$$

where $u = 2a\epsilon^{-2}$. Then

$$\mathbb{E}[\kappa] \leq \mathbb{E}[\kappa'] \leq 1 + \frac{2}{\epsilon^2} (a + \sqrt{\pi a} + 1).$$

Exercise 8.2. Consider the one-armed bandit problem: $\mathcal{E} = \{\mathcal{N}(\mu_1, 1) : \mu_1 \in \mathbb{R}\} \times \{\mathcal{N}(0, 1)\}$. Suppose that $\nu = (P_1, P_2) \in \mathcal{E}$, and P_1 has mean $\mu_1 = 1$. Evaluate

$$\limsup_{n \rightarrow \infty} \frac{R_n(\pi, \nu)}{\log n}$$

where π is the policy of asymptotically optimal UCB.

Exercise 8.3. Consider the setting of Exercise 8.2 and define a policy by

$$A_t = \begin{cases} 1 & \text{if } \hat{\mu}_1(t-1) + \sqrt{\frac{2 \log f(t)}{T_1(t-1)}} \geq 0, \\ 2 & \text{otherwise.} \end{cases}$$

Suppose that $\nu = (P_1, P_2)$ where $P_1 = \mathcal{N}(\mu_1, 1)$ and $P_2 = \mathcal{N}(0, 1)$. Prove that for the modified policy,

$$\limsup_{n \rightarrow \infty} \frac{R_n(\pi, \nu)}{\log n} \leq \begin{cases} 0 & \text{if } \mu_1 \geq 0, \\ -\frac{2}{\mu_1} & \text{if } \mu_1 < 0. \end{cases}$$

Theorem (Cramér). Suppose $X_1, \dots, X_n$ are i.i.d. random variables whose values lie in $\mathbb{R}^d$ with finite empirical mean defined by $S_n = \frac{1}{n} \sum_{i=1}^n X_i$. Define μ_n to be the law of S_n . Then μ_n satisfies a large deviation principle on $\mathbb{R}^d$ with rate function given by the Legendre transform of the cumulant generating function of X_1 . That is,

- For any closed set $F \subset \mathbb{R}^d$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mu_n(F) \leq - \inf_{x \in F} \Lambda^*(x),$$

- For any open set $G \subset \mathbb{R}^d$,

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mu_n(G) \geq - \inf_{x \in G} \Lambda^*(x),$$

$$\text{where } \Lambda^*(x) = \sup_{\lambda \in \mathbb{R}^d} \{\langle \lambda, x \rangle - \Lambda(\lambda)\} = \sup_{\lambda \in \mathbb{R}^d} \{\langle \lambda, x \rangle - \log \mathbb{E}[e^{\langle \lambda, X_1 \rangle}]\}.$$

Theorem (Gärtner-Ellis). Suppose $X_1, \dots, X_n$ are random variables whose values lie in $\mathbb{R}^d$ with finite empirical mean defined by $S_n = \frac{1}{n} \sum_{i=1}^n X_i$. Define μ_n to be the law of S_n . Suppose the limit

$$\Lambda(\lambda) := \lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}[e^{n\langle \lambda, S_n \rangle}]$$

exists for all $\lambda \in \mathbb{R}^d$ and $0 \in \mathcal{D}_\Lambda^\circ$, where $\mathcal{D}_\Lambda$ is the effective domain of Λ . Then

- For any closed set $F \subset \mathbb{R}^d$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mu_n(F) \leq - \inf_{x \in F} \Lambda^*(x),$$

- For any open set $G \subset \mathbb{R}^d$,

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mu_n(G) \geq - \inf_{x \in G \cap \mathcal{F}} \Lambda^*(x),$$

where $\Lambda^*(x)$ is the rate function given by the Legendre transform of $\Lambda(\lambda)$, and $\mathcal{F}$ is the set of exposed points of $\Lambda^*(x)$.

Furthermore, if Λ is essentially smooth and lower semicontinuous, then $\mathcal{F}$ can be replaced by the entire domain of Λ^* (i.e. LDP holds).